



Hassan Fayyaz Abbasi

Date of birth: 27/03/1999 | **Nationality:** Pakistani | **Phone number:**

(+92) 3135189502 (Mobile) | **Email address:** mhassanabbasi313@gmail.com | **Website:**

<https://hassan-abbasi-dev.vercel.app/> | **LinkedIn:** www.linkedin.com/in/hassan-abbasi-dev |

Address: House no 894 dhoke kasmiryan, 46000, Rawalpindi, Pakistan (Home)

● ABOUT ME

Software Engineer with 5+ years of experience bridging Full Stack Development, AI/LLM integration, and Geospatial (GIS) systems. I specialize in transforming complex AI models and enterprise ERPs into secure, production-grade applications. From architecting multi-agent AI workflows and integrating SAP financial modules to building custom GIS mapping tools, my focus is on writing clean, scalable code that solves tangible business problems.

● WORK EXPERIENCE

AI / LLM ENGINEER – PROPMAX.AI – 01/03/2025 – Current – ISLAMABAD, PAKISTAN

- Designed and developed AI-powered **chatbot and voice assistant systems** (using Whisper) to automate customer interactions, reducing routine support ticket volume by **~40%**.
- Implemented **Retrieval-Augmented Generation (RAG)** pipelines for intelligent querying over **CRM, ERP, and internal datasets**, improving data retrieval accuracy by **35%** through hybrid search optimization.
- Built scalable AI applications using **Python and LangChain**, developing **agent-based** architectures with function calling to connect LLMs directly to **5+ internal REST APIs** for autonomous task execution.
- Engineered **advanced prompts** using few-shot learning and strict system constraints, pushing structured JSON output compliance from **~70% to 95%** for seamless, bug-free frontend integration.
- Optimized LLM inference by implementing **semantic caching and model routing**, **cutting average API latency by 45% (from ~4s down to 2s)** and reducing monthly cloud costs by 30%.
- Designed multi-step reasoning workflows to handle **complex business logic**, **integrating AI systems** directly with **PostgreSQL databases for real-time**, factual data operations.
- Implemented strict **Role-Based Access Control (RBAC)** at the backend, ensuring **100%** data isolation so **Customers, Advisors, and Branch Managers** could only query financial data or generate documents (**like KYC/KFI**) they were explicitly authorized to access.

TECHNICAL TEAM LEAD (PART-TIME) – SIRAFY DIGITAL SOLUTIONS – 01/05/2024 – Current – RAWALPINDI, PAKISTAN

- Led a team of developers in designing and delivering **web and enterprise applications**, ensuring high-quality code and timely delivery
- Managed **task planning, code reviews, and sprint coordination** in an Agile environment
- Provided technical guidance on **architecture, scalability, and best practices** across projects
- Oversaw development of **CRM, ERP, and custom web solutions**, ensuring performance and reliability
- Collaborated with stakeholders to gather requirements and translate them into **technical solutions**
- Mentored junior developers, improving team productivity and code quality

FULLSTACK SOFTWARE ENGINEER – PROPSURE DIGITAL SOLUTIONS – 03/09/2021 – 28/02/2025 – ISLAMABAD, PAKISTAN

- **Architected and maintained** scalable PropTech applications using **React.js, Laravel, and Node.js**, supporting a platform with over **10,000 active** users and complex property management workflows.
- **Led the complex bidirectional integration** between our local **PropTech ERP and SAP/Haball**. Restructured the core database schema and built a custom Laravel data transformation layer to ensure **financial booking data synced perfectly**, eliminating manual reconciliation and ensuring **100% accuracy in financial reporting**.
- Engineered an advanced **GIS module replicating QGIS functionality directly in the browser**; integrated **Turf.js** on the frontend and **PostGIS** on the backend to enable precise **plot division, merging, and boundary mapping with zero geometric errors**.

- Optimized **PostgreSQL** and **MySQL** database schemas through advanced **indexing**, reducing complex page **load times by 35%** and ensuring a smooth experience for users handling large datasets.
- Integrated various **3rd-party APIs and web services**, ensuring seamless **data synchronization across external enterprise systems** and internal modules.

FULLSTACK DVELOPER – ORCALO HOLDINGS – 01/06/2022 – 31/07/2023 – RAWALPINDI, PAKISTAN

- Designed and documented highly scalable **RESTful APIs using Swagger/OpenAPI**, creating a unified reference point that drastically reduced onboarding time for new frontend developers.\
- Streamlined the development lifecycle by introducing **automated testing and standardized API contracts**, **improving cross-team collaboration** and system reliability.
- Applied strong **problem-solving** skills to troubleshoot **complex API design challenges**, contributing to a **99.9%** uptime and stability for our core web applications.

FREELANCE WEB DEVELOPER – FREELANCE – 05/05/2020 – Current – PAKISTAN

- Optimized e-commerce platforms by improving navigation and **integrating secure payment gateways**, directly resulting in a **25% increase in sales** conversion rates for clients.
- Developed a real-time Daily Sales Management System and built dynamic **Power BI dashboards**, giving stakeholders clear, **data-driven insights** for strategic decision-making.
- Applied **GIS technology** for location-based problem-solving in local **real estate and logistics** projects, helping clients **visualize spatial data** to optimize their operations.

REACT NATIVE INTERN – GRAANA – 11/08/2021 – 31/10/2021 – RAWALPINDI, PAKISTAN

- Developed and deployed **cross-platform** mobile applications using **React Native**, significantly reducing development time for **iOS and Android platforms** through high code reusability.

● **EDUCATION AND TRAINING**

15/02/2019 – 30/11/2021 Rawalpindi, Pakistan
MASTERS IN COMPUTER SCIENCE Barani Institute Of Technology

Computer Programming, Data Structures and Algorithms, Database Management, Operating Systems, Networking and Security, Software Engineering, Machine Learning and AI, Cloud Computing, Cybersecurity, Big Data, Human-Computer Interaction (HCI), Ethics in Computing, Research and Innovation, Soft Skills, Capstone Project.

Website <http://www.biit.edu.pk/>

2015 – 2017 Islamabad, Pakistan
HIGHER SECONDARY SCHOOL CERTIFICATE (ICS) Islamabad College For Boys

Website <https://icb.edu.pk/>

2012 – 2014 Islamabad, Pakistan
SECONDARY SCHOOL CERTIFICATE IN SCIENCE Islamabad College For Boys G-6/3 Islamabad

Website <https://icb.edu.pk/>

● **LANGUAGE SKILLS**

Mother tongue(s): **URDU**
 Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● SKILLS

AI & LLM Engineering

LangChain | RAG Pipelines | Prompt Engineering | Whisper (Audio) | Function Calling | Autonomous Agents | LLM Evaluation (LangSmith)

Backend

Laravel | Python | Node | JavaScript | fastapi | Haball Integration | API Integrations | Postman | Google Maps API

FrontEnd

Bootstrap | React, Next.js, React Native | REST | SAP Integrations | JSON | ESRI APIs | HTML | Leaflet

GIS & Geospatial

GeoServer | PostGIS | GDAL | Geospatial: (Geopandas, GDAL, arcpy) | Turf.js | OpenLayers | GeoPandas | HTML PHP CSS JS (leaflet Open layers Google ESRI APIs)

DBMS/Spatial Databases

PostgreSQL | SQL Server | MySQL | mongodb

Cloud Computing

S3 bitbucket | Atlassian (Bitbucket, Confluence, Jira) | Git (Github / Gitlab) | Elastic search | Unit Testing | BI tools and applications (Amazon AWS)

● PROJECTS

Aydai

Enterprise Role-Based AI Assistant: Engineered a comprehensive, role-aware AI assistant that goes far beyond simple customer support. By integrating strict backend Role-Based Access Control (RBAC) with LangChain function calling, the system dynamically adapts to the user: Customers can securely query their personal booking and financial data; Advisors can generate KYC, KFI documents, and contracts or check available products; Branch Managers can monitor team metrics; and Admins maintain full system oversight. This architecture guaranteed 100% data isolation, ensuring users only ever interact with the data and tools they are explicitly authorized to access.

Capstone

The capstone project I contributed to is a comprehensive real estate management system that incorporates inventory tracking, booking, installment payments, rental management, buyback transactions, and SAP finance integration.

- Capstone (Real Estate & SAP Finance): Integrated SAP finance modules for a comprehensive real estate ERP, automating booking, inventory, and installment workflows for 500+ properties while ensuring strict compliance with financial reporting standards.

Prop Feasibility

In my feasibility project, I utilized mapping technology to deliver comprehensive information about customer locations, including road networks, schools, brands, mosques, and more. I integrated Geo server to generate interactive layers, allowing users to access distance and time data for various destinations. This approach enhanced decision-making by providing valuable location-based insights and optimizing route planning.

Online Property Verification System(OPVS)

I worked on an online property verification project that generates comprehensive reports, ensuring legal compliance and ownership authenticity. It integrates with databases, conducts encumbrance checks, verifies property documents, and offers a user-friendly interface. Specifically designed for use with the Capital Development Authority (CDA), it streamlines property transactions while maintaining data security and staying up-to-date with regulations.

Chatbot

I also contributed to the development of a chatbot aimed at reducing the workload of our customer support agents. The chatbot effectively handles routine inquiries, offers quick solutions, and directs complex issues to human agents when necessary. This automation not only enhances customer service but also frees up valuable agent time for more complex tasks, ultimately improving efficiency and customer satisfaction.

Propsure Erp

In the Propsure ERP project, we implemented an advanced GIS (Geographic Information System) module that replicated QGIS functionality. This module allowed us to efficiently manage land parcels, dividing and merging plots, and defining property boundaries. Key features included:

Plot Division and Merging: Users could easily subdivide or combine land parcels, enabling flexible property management.

Boundary Definition: Precise boundary delineation tools facilitated accurate property mapping.

Enhanced GIS Capabilities: We integrated powerful GIS capabilities within the ERP system, streamlining land management tasks and ensuring data accuracy.

Efficiency Boost: This functionality significantly improved efficiency in land and property management, reducing manual effort and minimizing errors.

Other Project:

PropSure Website (<https://propsure.com.pk/>)

PropRetailer (<https://propmarker.propsure.rocks/>)

workSap (<https://worksapp.com.pk/>)

Propsure Mobile App

Aluminium Website (<https://waqaraluminium.com/>)

Api Swaggers

Gypsum Board(<https://gypsumboarddecoralriaz.com/>)

and many more..